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1 HTML Essentials

Basic document structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Page Title</title>
</head>
<body>
  <!-- your content -->
</body>
</html>
```

Common tags

Tag	Use
<h1> ... <h6>	Headings (1 = most important, use only one <h1> per page)
<p>	Paragraph of text
	Link to another page or section
	Image — alt describes it for screen readers & SEO
/ /	Unordered / ordered list / list item
<div>	Generic block-level container
	Generic inline container
<section>, <nav>, <header>, <footer>, <article>	Semantic layout tags — better for SEO & accessibility than a page full of <div>s
<form>, <input>, <button>	Collect user input

2 CSS Essentials

Selectors

```
.class      { } /* selects elements with class="class"
*/ #id      { } /* selects the element with id="id"
*/ element  { } /* selects every <element> */ a, b
{ } /* selects a AND b */ parent > child{ } /* direct
child only */ a:hover { } /* selects a on mouse
hover */
```

The box model (outside → in)

margin → border → padding → content

Flexbox (1-D layout)

```
.row {
  display: flex;
  justify-content: center; /* main axis: start | center | space-between */
  align-items: center;     /* cross axis: start | center | stretch */
  gap: 1rem;
}
```

Grid (2-D layout)

```
.grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 1rem;
}
```

Properties you'll use constantly

Property	Example
color / background	color: #142433; background: #F4F6F3;
font-size / font-weight	font-size: 1.1rem; font-weight: 600;
margin / padding	margin: 0 auto; padding: 1rem 1.5rem;
border-radius	border-radius: 12px;
box-shadow	box-shadow: 0 10px 20px rgba(0,0,0,.15);
transition	transition: transform 0.25s ease;

3 JavaScript Essentials

Variables & functions

```
const name = "Neil"; // won't be reassigned
let score = 0;       // can be reassigned

function greet(name) { return "Hi, " + name; }
const greet2 = (name) => "Hi, " + name; // arrow function
```

Selecting & changing the page (DOM)

```
const btn = document.querySelector('.my-button');
const all = document.querySelectorAll('.card'); // returns a list

btn.textContent = "Clicked!";
btn.classList.add('is-active');
btn.style.color = "teal";
```

Listening for events

```
btn.addEventListener('click', function () {
  console.log('Button was clicked');
});
```

Arrays & objects

```
const skills = ["HTML", "CSS", "JavaScript"];
skills.push("SQL"); // add to end
skills.forEach(s => console.log(s)); // loop through
const upper = skills.map(s => s.toUpperCase());
const student = { name: "Neil", year: 3 };
console.log(student.name); // dot notation
console.log(student["year"]); // bracket notation
```

4 Quick Tips & Common Gotchas

- Always close your tags — an unclosed `<div>` can break your whole layout.
- Use semantic HTML (`<nav>`, `<section>`, `<footer>`...) — it helps SEO and screen readers.
- Design mobile-first, then add a media query for larger screens: `@media (min-width: 768px) { ... }`
- Every `<img>` needs meaningful alt text — describe the image, not just "photo."
- Prefer `const` by default, let only when a value changes, and avoid `var` entirely.
- `console.log()` liberally while debugging — then remove or comment out before submitting.
- Validate your HTML/CSS with the W3C Validator if something looks broken and you can't tell why.

5 Handy References

- MDN Web Docs — developer.mozilla.org (the best HTML/CSS/JS reference, period)
- Can I Use — caniuse.com (check browser support before using a new CSS/JS feature)
- freeCodeCamp — freecodecamp.org (free structured practice)
- W3C Validator — validator.w3.org (catch HTML mistakes)